

Unit 4, Lesson 4: Evaluating Numerical Expressions with Integer Exponents – Part 1

Benchmark

MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.

Learning Target

- ☐ I can evaluate a numerical expression using the laws of exponents.

4.4.1 Warm-Up: Bell Work

1.
- For each expression, apply one or more exponent laws to write an equivalent expression with a rational base and positive exponent.

Given Expression	Equivalent Expression
$\left(\frac{4}{5}\right)^{-5} \left(\frac{4}{5}\right)^2$	
$(3^4)^{-6}$	
$\frac{7^8}{7^6}$	
$\left(\frac{12}{5}\right)^{-4}$	

4.4.2 Guided Instruction: Using the Laws of Exponents to Evaluate Expressions

1. Melanie and Adonis each evaluated the expression $\frac{(5)^{-4}(5)^3}{5}$. Each student's steps are shown.

Melanie	Adonis
$\frac{(5)^{-4}(5)^3}{5}$ $= \frac{5^9}{5}$ $= 5^2$ $= 25$	$\frac{(5)^{-4}(5)^3}{5}$ $= \frac{\frac{1}{5^4} \cdot (5)^3}{5}$ $= \frac{\frac{1}{625} \cdot 125}{5}$ $= \frac{125}{5}$ $= 25$

Discuss with your partner how Melanie's steps and Adonis' steps are similar and different from each other. Then, summarize your discussion in the table.

Similarities	Differences



When evaluating exponential expressions, it is helpful to apply the laws of exponents before evaluating.

2. Complete the steps to evaluate the expression $\frac{6^3 \cdot 4}{6^2}$ by completing the description of each step and writing the resulting equivalent expression on the line that follows.

a. Apply

- ☐ product of power law.
- ☐ power of a power law.

b. Apply

- ☐ quotient of power law.
- ☐ power of a power law.

c. Evaluate.

3. Consider the expression $\left(\frac{4}{5}\right)^{-2} \cdot (4)^2$.

a. Discuss with your partner which laws of exponents could be used to evaluate the expression.

b. Evaluate $\left(\frac{4}{5}\right)^{-2} \cdot (4)^2$.

4.4.3 Try It: Evaluating Expressions

1. Evaluate each expression. Show your work in the space provided. Then, check off the exponent laws applied to each expression from the list provided.

a. $\left(\frac{2}{4}\right)^{-2} - 3$

Exponent Laws Applied	
☐	identity exponent
☐	negative exponent
☐	power of a power
☐	power of a product
☐	power of a quotient
☐	product of powers
☐	quotient of powers
☐	zero exponent

b. $\frac{(-7)^3}{(-7)^0(-7)^2}$

Exponent Laws Applied	
☐	identity exponent
☐	negative exponent
☐	power of a power
☐	power of a product
☐	power of a quotient
☐	product of powers
☐	quotient of powers
☐	zero exponent

4.4.4 Activity: Card Sort

1. Work with your partner to match each expression card with the exponent law and value cards. The exponent law cards indicate which laws of exponents can be applied to the expression before evaluating and the value cards indicate the value of the expression. Write the letter or number associated with the matching cards in the table.

Card	Expression	Value	Exponent Law(s)
a.	$\frac{1}{4^{-2}} - \frac{1}{4^4}$		
b.	$\frac{(-2)^0(-2)^4}{(-2)^{-8}}$		
c.	$\frac{\left(\frac{3}{5}\right)^2}{\left(\frac{3}{5}\right)^{-1}}$		
d.	$\frac{(-15)^2}{(-15)^{-4}(-15)^4}$		
e.	$3^2 \cdot \left(\frac{2}{3}\right)^{-1}$		

4.4.5 Your Turn!

1. Apply the laws of exponents to evaluate each expression. Show your steps in the process.

a. $\frac{(0.6)^{-2} \cdot (0.6)^7}{(0.6)^3}$

b. $(-4)^2 \cdot \left(\frac{-5}{4}\right)^{-4}$

2. Cristian was evaluating the expression $\left(\frac{5}{6}\right)^{-5} \cdot 6^2$.

- a. Circle the expressions that are equivalent to $\left(\frac{5}{6}\right)^{-5} \cdot 6^2$.

$$\frac{5^{-5}}{6^{-5}} \cdot 6^2$$

$$5^{-5} \cdot \frac{6^2}{6}$$

$$\frac{5^{-5}}{6^{-5}} \cdot 6^2$$

$$\left(\frac{6}{5}\right)^5 \cdot 6^2$$

- b. Write another equivalent expression.

4.4.6 Wrap-Up: Error Analysis

1. Two students each evaluated an expression. Their work is shown in the table. Each student made a mistake. Circle the step where each student made an error and describe the mistake they made.

Problem	Describe the Mistake
$\left(\frac{1}{5}\right)^4 \cdot \left(\frac{1}{5}\right)^{-2} \cdot 5^{-1}$ $= \left(\frac{1}{5}\right)^{-8} \cdot 5^{-1}$ $= 5^8 \cdot 5^{-1}$ $= 5^7$ $= 78,125$	
$\left(\frac{3}{2}\right)^{-3} \cdot (2)^3$ $= \frac{2^{-3}}{3^{-3}} \cdot 2^3$ $= \frac{2^0}{3^{-3}}$ $= \frac{1}{3^{-3}}$ $= 3^3$ $= 27$	